time.sim

Purpose

`time.sim` sets the simulation start date, end date, and the sub-daily time step. All time-driven processes (climate reads, routing, output aggregation) use these values.

Source

- Reader: `src/time_read.f90`
- Type definition: `src/time_module.f90`, `type time_current`

Format

- Line 1: title line. Skipped.
- Line 2: header line. Skipped.
- Line 3: five integer values, list-directed.

#	Name	Type	Description
1	day_start	int	Julian day of year on which the simulation starts (1 to 366)
2	yrc_start	int	Calendar year on which the simulation starts
3	day_end	int	Julian day of year on which the simulation ends
4	yrc_end	int	Calendar year on which the simulation ends
5	step	int	Sub-daily time step code (see below)

Sub-daily time step values

The type definition in `time_module.f90` documents these step codes:

Value	Meaning
0	daily
1	12-hour increment
24	hourly
96	15 minutes
1440	per minute

Defaults and derived values

The reader applies these adjustments after parsing:

- `step <= 0` is forced to `1`.
- `day_start <= 0` is forced to `1`.
- `time%nbyr` is set to `yrc_end - yrc_start + 1`.
- `time%mo`, `time%day_mo`, and `time%mo_start` are computed from `day_start` via the `xmon` helper.
- `time%yrc` is initialised to `yrc_start`.

Example

`refdata/Ames_sub1/time.sim`:

```
time.sim:
day_start  yrc_start   day_end   yrc_end      step
       0      1975          0      2020         0
```

This runs from 1 Jan 1975 through 31 Dec 2020 (the zero values for `day_start` and `day_end` are interpreted as the first and last day of the year). The `step` value of `0` is forced to `1` by the reader.

Related files

- `print.prt`. Controls which outputs are written within this simulation window.